

SUBMISSION_1

Our solution trains the GPT2-large model on the provided training data, with a twist:

- We synthesize self-built novel additional family data for further training for our model
- These results come from training on 12 families, testing on 3 unseen families:

0.722

OOD3 Top-1

family-wise avg

0.999

OOD3 Top-5

family-wise avg

0.847

OOD3 MRR

family-wise avg

15

Families

12 train / 3 test

Submission surface

-  nextstep.csv
-  completion.csv
-  anomaly.csv
-  outputs/gpt_large_train12
-  n-gram fallback

The Data Problem

- With most model solutions, performance drops when tested on unseen families
- It is hard to distinguish if models learn the process grammar or if they memorization the grammar of the families
- All baseline performance scores are not satisfactory for semiconductor chip production

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"STEP","DESCRIPTION"
"RECEIVE WAFER LOT","Receive the incoming wafer lot and verify lot ID, product type, and route version."
"LOT IDENTIFICATION","Confirm wafer count, traveler information, and manufacturing order consistency."
"INITIAL WAFER INSPECTION","Inspect wafers for visible defects, handling damage, contamination, and edge chips."
"MEASURE THICKNESS","Measure incoming wafer thickness at center and edge points."
"MEASURE SURFACE PARTICLES","Perform particle scan to assess surface cleanliness before processing."
"PRE CLEAN WAFER","Remove loose particles and residues prior to wet processing."
"RCA CLEAN 1","Apply alkaline clean to remove organic contamination; monitor temperature and time."
"RCA CLEAN 2","Apply acidic clean to remove metallic residues and remaining contamination."
"HF DIP","Remove native oxide with dilute HF and prepare a clean silicon surface."
"DRY WAFER","Dry wafers using spin-rinse-dry or nitrogen drying."
"SUBSTRATE CHECK","Verify substrate orientation, resistivity, and starting material specification."
"EPITAXY PREP","Prepare substrate surface for epitaxial growth with a final pre-clean."
"EPITAXIAL DEPOSITION","Grow the epitaxial silicon layer; control temperature, gas flow, and deposition rate."
"MEASURE EPITAXY THICKNESS","Measure epitaxial layer thickness and uniformity across the wafer."
"MEASURE RESISTIVITY","Measure resistivity or sheet resistance to confirm layer properties."
"EPITAXY ANNEAL","Anneal the epitaxial layer to improve crystal quality and reduce defects."
"THERMAL OXIDATION","Grow a thin oxide layer for masking or interface preparation."
"MEASURE OXIDE THICKNESS","Measure oxide thickness and uniformity by metrology."
"SPIN COAT PHOTORESIST","Coat photoresist; track spin speed, thickness, and coating quality."
"SOFT BAKE","Remove solvent from the resist film and stabilize it for exposure."
"ALIGN MASK LEVEL 1","Align the first mask for junction or window formation."
"EXPOSE LITHO LEVEL 1","Expose the resist using the first mask; record dose and focus."
"DEVELOP PHOTORESIST","Develop exposed resist to define the first pattern."
"PATTERN INSPECTION LEVEL 1","Inspect pattern fidelity, overlay, and defect density."
"OXIDE ETCH","Etch oxide in opened regions to create the required windows."
"STRIP PHOTORESIST","Remove resist after etch."
"CLEAN AFTER ETCH","Clean residues and particles after the first etch step."
"MEASURE OPENING CD","Measure critical dimension of the etched openings."
"IMPLANT P BODY","Implant p-type body region with controlled dose and energy."
```

h12014883, 2 days ago • Added cases

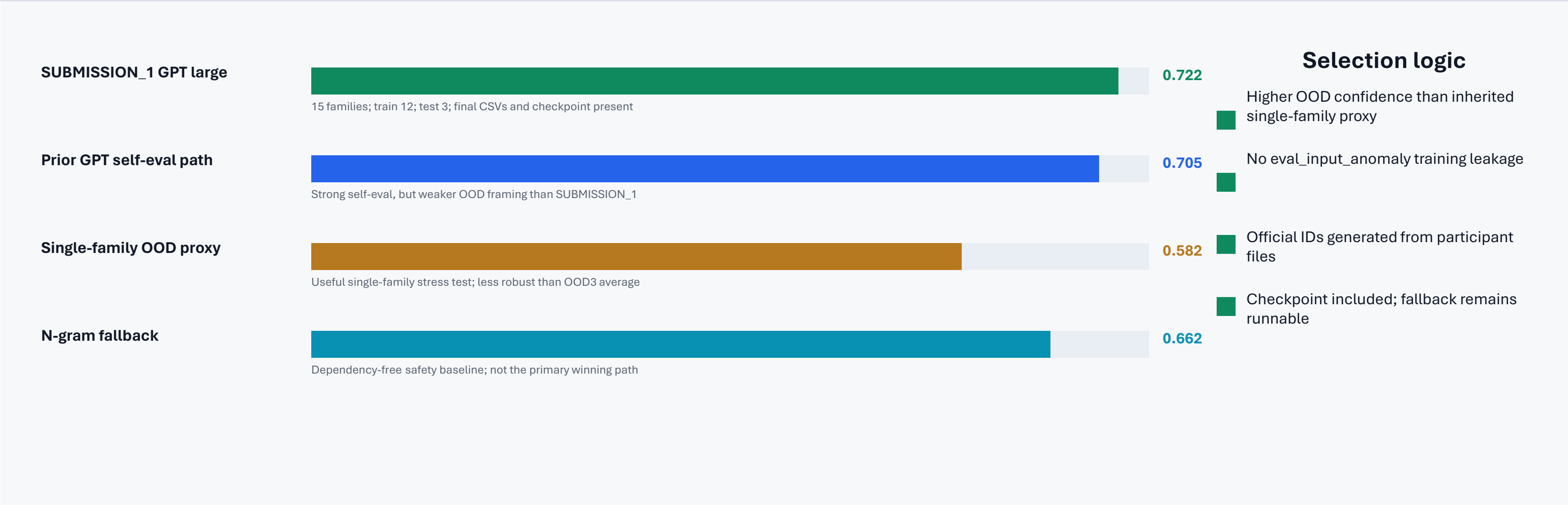
Score exactly where the jury scores

Each official task has a generated CSV, a legal source path, and an auditable local proof.

Jury task	Submitted file	What it scores	SUBMISSION_1 proof	Risk
1. Next step	nextstep.csv	Top-1 / Top-3 / Top-5 / MRR	GPT large OOD3 Top-1 0.722	Low
2. Completion	completion.csv	Normalized edit distance plus exact/token/block	Same checkpoint, official IDs	Medium
3. Anomaly	anomaly.csv	Validity, score, violated rule name	Rule validator, official rule strings	Low
Legality	repo evidence	No hidden eval training	Official eval only used for inference	Low
Reproducibility	REPORT + README	Run path and dependencies	Checkpoint + n-gram fallback	Low

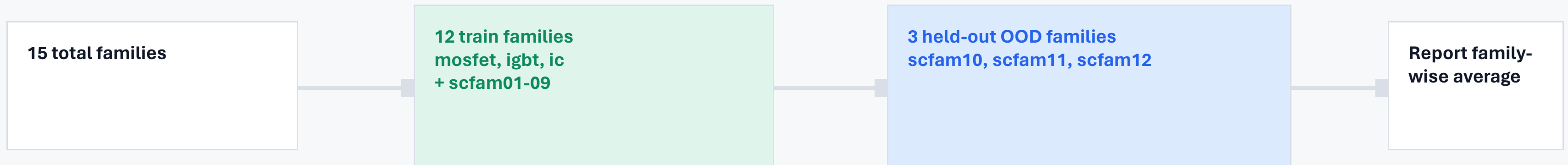
Why this branch ranks first

The best winning chance is the strongest legal OOD result that also ships clean official-format artifacts.



OOD protocol is explicit and judge-aligned

SUBMISSION_1 tests whether the model learned process grammar, not just known family routes.



Why this matters

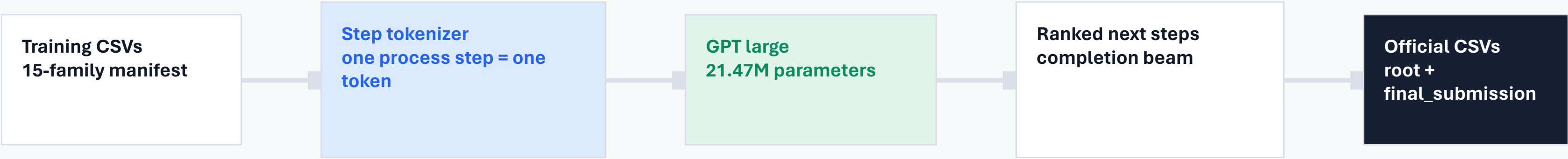
A hidden fourth family rewards transfer. The local score must therefore penalize family memorization before the final submission is frozen.

Legal boundary

Official eval_input_valid.csv and eval_input_anomaly.csv are used only to produce final CSVs, not as training data.

System design stays simple enough to trust

The model improves sequence predictions; the validator owns hard rule compliance.



Fallback path

Trigram n-gram runs without torch and can regenerate official-shaped CSVs from a clean checkout.

Safety path

GPT runner refuses random-init submissions unless explicitly overridden, preventing silent bad CSVs.

Evidence: GPT beats the safe baseline on OOD3

The measured lift is on the train-12/test-3 protocol, not on the final hidden eval files.



+0.058 Top-1 lift vs n-gram

Family-wise OOD3 average: Top-1 0.7216, Top-5 0.9989, MRR 0.8474

Blockers closed before final polish

The package now fails closed on format, legality, and checkpoint availability.



Official IDs

Root CSVs use valid_0001... and anomaly_... copied through from participant inputs.



Checkpoint present

outputs/gpt_large_train12 contains config, generation config, and model.safetensors.



Required docs

REPORT.md, README.md, LICENSE, requirements.txt are present at repo root.



Legal usage

Official eval inputs are used only for final inference; not for training.



Fallback

n-gram official bundle remains available if checkpoint transfer fails.

Final files

[nextstep.csv](#)

[completion.csv](#)

[anomaly.csv](#)